

Battery Drain Experiment — Project Proposal

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Experiment Description

I will investigate how four common iPhone settings affect **battery drain rate**. Smartphones are integral to daily life, and understanding which settings most impact battery consumption has practical value. By systematically varying brightness, WiFi connectivity, app workload, and Low Power Mode, I can quantify main effects and interactions between these factors.

Response Variable

Battery percentage drop (%) over a fixed 15-minute interval. Each run begins by recording the current battery level, applying the treatment settings, waiting 15 minutes, and recording the final level. The response is the difference (start % – end %).

Factors and Levels

Factor	Low (–)	High (+)
A: Screen Brightness	25%	100%
B: WiFi	Off (Airplane mode)	On (connected)
C: App Workload	Idle (home screen, no apps)	Active (local video on loop)
D: Low Power Mode	On	Off

All four factors are independently controllable with no logical conflicts. Factor C uses a pre-downloaded video (not streaming), so it does not depend on WiFi status.

Statistical Design

I will use a 2^4 **full factorial design** (Chapter 4) with $n = 3$ replicates per treatment combination, giving $16 \times 3 = 48$ total runs. Each run takes 15 minutes, so the experiment is feasible over 2–3 days. Runs will be randomized within each day, with **Day** as a blocking factor to account for day-to-day variation.

Planned Statistical Analysis

- Full factorial ANOVA with all main effects and two-factor interactions, plus Day as a blocking factor
- Half-normal plot of estimated effects to identify significant factors
- Interaction plots for significant two-factor interactions
- Residual diagnostics: normal Q-Q plot, residuals vs. fitted values, and residuals vs. run order to check normality, constant variance, and independence
- Multiple comparisons (Tukey or Bonferroni) if needed
- Location-dispersion modeling using the 3 replicates per cell to examine whether certain settings also affect variability in battery drain